| Question | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| 2 | *(a)* | (i) | Kinetic energy and [gravitational] potential energy referred to (1)<br>$\frac{1}{2}mv^2 - \frac{GMm}{r}$ = total [initial] energy **or** ΔKE = [–]ΔPE (1)<br>Final and initial energy are zero and equal [accept final KE = 0 or final PE = 0] (1) | 3 | | | 3 | | |
| | | (ii) | Rearrangement / simplification i.e. $v^2 = \frac{2GM}{r}$ (1) [accept *m* = 1 inserted]<br>Substitution e.g. $v^2 = \frac{2\times 6.67\times 10^{-11}\times 1.99\times 10^{30}}{6.96\times 10^{8}}$ (1)<br>Answer = 618 [or 620] km $s^{-1}$ (1) c.a.o. [accept ~600 km $s^{-1}$ b.o.d.] | 1 | 1<br>1 | | 3 | 3 | |
| | *(b)* | (i) | Applying KE of molecule/atom/particle = $\frac{3}{2}kT$ (or deriving) (1)<br>Rearrangement e.g. accept $v^2 = \frac{3kT}{m}$ (1)<br>Substitution e.g. $v^2 = \frac{3\times 1.38\times 10^{-23}\times 5780}{9.11\times 10^{-31}}$ (1)<br>Answer = 513 km/s (1)<br>[accept just ~500 km $s^{-1}$ only with correct substitution] | 1 | 1<br>1<br>1 | | 4 | 3 | |
| | | (ii) | Many electrons have enough KE to escape **(ecf)** or just 'some electrons escape' (1)<br>Because (b)(i) close to (a)(ii) / some electrons have higher velocity / [Boltzmann] distribution / collisions] **(ecf)** (1)<br>Or protons don't escape | | 2 | | 2 | | |
| | | (iii) | Valid method employed e.g. electrostatic force & gravitational forces calculated/equated or other (1)<br>Valid calculation carried out correctly e.g. $F_E = 2.4\times 10^{-28}$ N **and** $F_g = 2.5\times 10^{-28}$ N or charge = 0.084 C (1)<br>[Or using *GMm* = *kQq* → 1.2 × $10^{-10}$ [N $m^2$] and 1.15 × $10^{-10}$ [N $m^2$]]<br>Valid conclusion (not independent) e.g. she's quite close (1) **ecf** from calculation of forces [not fields] or alternative above.<br>[Accept: *E* force bigger / *G* force smaller] | | | 3 | 3 | 2 | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | (iv) | Sun electrons ≈ $\frac{1.99\times10^{30}}{1.66\times10^{-27}}$ ≈ $1.2\times10^{57}$ (assumption dependent, allow Sun composed of deuterium) (1) [Accept $M_S$ / 1 u]<br>$\frac{0.08}{e}$ = $5\times10^{17}$ electrons lost **or** charge on Sun's electrons = $1.92\times10^{38}$ [C] (1)<br>% lost ≈ $10^{-38}$ (1) | | 3 | | 3 | 3 | |
| | | | **Question 2 total** | **5** | **10** | **3** | **18** | **11** | **0** |